

7.3. 7.8

7. 矩阵特征值是 2 -1 -4

∴ 最大值是 2, 求出特征向量 $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

8. 矩阵最大特征值是 9 ∴ 归一化为 $\begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$

二次型矩阵为 $A = \begin{bmatrix} 7 & -4 & -2 \\ -4 & 1 & -4 \\ -2 & -4 & 7 \end{bmatrix}$

求特征向量

$$A - 9I: \begin{bmatrix} -2 & -4 & -2 \\ -4 & -8 & -4 \\ -2 & -4 & -2 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad x_1 = -2x_2 - x_3$$
$$x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

∴ 一个特征向量为 $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$, 归一化 $\begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}$

7.4. 12. 17. 23

$$^{12}A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = (2 - \lambda)(3 - \lambda) = 0$$

$$\lambda = 2 \text{ or } \lambda = 3$$

当 $\lambda = 3$ 对应特征向量 $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\lambda = 2$ 对应特征向量 $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\therefore V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad b_1 = \sqrt{3} \quad b_2 = \sqrt{2} \quad \Sigma = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}$$

$$\mu_1 = \frac{1}{b_1} A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$\mu_2 = \frac{1}{b_2} A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

$$\text{由提示, } \mu_3 = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \quad U = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix}$$

$$\therefore A = U \Sigma V^T = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

1) "正交矩阵行列式为 ± 1 " 若 A 为正交矩阵
 $\det A^T A = \det I = (\det A)^2 = 1 \therefore \det A = \pm 1$

$\therefore A = U \Sigma V^T$ 其中 U 和 V^T 为正交矩阵

$$\therefore |\det A| = |\det U \Sigma V^T| = |\det U \cdot \det \Sigma \cdot \det V^T| \\ = |\det \Sigma|$$

" Σ 为对角矩阵, $\therefore \det \Sigma = \sigma_1 \cdot \sigma_2 \cdots \sigma_n$ "

$\therefore$ 命题得证.

$$2) U \Sigma = [\sigma_1 u_1 \quad \sigma_2 u_2 \quad \sigma_3 u_3 \cdots \sigma_r u_r \quad 0 \cdots 0]$$

$$(U \Sigma) V^T = [\sigma_1 u_1 \quad \sigma_2 u_2 \cdots \sigma_r u_r \quad 0 \cdots 0] \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix}$$

$$= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots \sigma_r u_r v_r^T = A$$

证毕.

8. 1. 7. 11. 12. 21

7. 令 $A = [v_1 \ v_2 \ v_3]$ 求

$$[A \ p_1 \ p_2 \ p_3]$$

行化简得:

$$\begin{bmatrix} 1 & 0 & 0 & 2 & 2 & 2 \\ 0 & 1 & 0 & 1 & -4 & 2 \\ 0 & 0 & 1 & -1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 & -5 \end{bmatrix}$$

a. $p_1 = 2v_1 + v_2 - v_3$ $\therefore p_1$ 属于 $\text{span } S$

" $2 + 1 - 1 \neq 1$ $\therefore p_1$ 不属于 $\text{att } S$

b. $p_2 = 2v_1 - 4v_2 + 3v_3$ $\therefore p_2$ 属于 $\text{span } S$

" $2 - 4 + 3 = 1$ $\therefore p_2$ 也属于 $\text{att } S$

c. $p_3 \notin \text{span } S$ 因为 $0 \neq -5$, 无解.

$\therefore p_3$ 也不属于 $\text{att } S$

11 $a \checkmark$

b. X 权重之和要为 1

c $\checkmark$

d X 平面是子空间的平移

e $\checkmark$

12. $a \quad X \text{ aff } S = \{X\}$

b $\checkmark$

c $\checkmark$

d X 仅当其归属于 R^3 才能说二维平面是超平面

e. $\checkmark$

2) $A \subset B$

" B 是仿射的, $\therefore B$ 包含所有仿射组合,

" $A \subset B$ $\therefore B$ 肯定包含 A 的所有仿射组合

$\therefore \text{aff } A \subset B$

①

$$S = \{\vec{v}_1 \cdots \vec{v}_k\}$$

$$\forall \vec{p} \in \text{aff } S$$

$$\vec{p} = c_1 \vec{v}_1 + \cdots + c_k \vec{v}_k \quad (c_1 + \cdots + c_k = 1)$$

$c_1 \cdots c_k$ 是唯一的

$$\textcircled{2} \quad c_1 \vec{v}_1 + \cdots + c_k \vec{v}_k = \vec{0} \quad \text{当且仅当 } c_1 = \cdots = c_k = 0$$

证明①②命题等价

$$\textcircled{2} \Leftrightarrow \vec{v}_1 \cdots \vec{v}_k \text{ 线性无关, 记为 } \textcircled{3}$$

若③成立则①自然成立 $\textcircled{3} \Rightarrow \textcircled{1}$

同时若①成立则 $[\vec{v}_1 \cdots \vec{v}_k] x = \vec{p}$

仅有唯一解. 此时说明 $\vec{v}_1 \cdots \vec{v}_k$ 线性无关

$$\therefore \textcircled{1} \Rightarrow \textcircled{3} \quad \therefore \textcircled{3} \Leftrightarrow \textcircled{1}$$

$$\therefore \textcircled{2} \Leftrightarrow \textcircled{3} \quad \textcircled{3} \Leftrightarrow \textcircled{1} \quad \therefore \textcircled{2} \Leftrightarrow \textcircled{1}$$

证毕.